

# OLIVIER BOUËT-WILLAUMEZ

PhD Student ~ Computer Science

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## CURRENT POSITION

I am currently a PhD student in Computer Science at the University of Paris-Est Créteil (UPEC), in the Laboratory of Algorithms, Complexity, and Logic (LACL), under the supervision of Nihal Pekergin, Adrien Le Coënt and Benoît Barbot. My thesis is titled: "Machine Learning of Continuous Systems Based on Probabilistic Approximations."

## SKILLS

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|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Mathematics (Theoretical and Applied)</b> | Machine Learning, High-Dimensional Statistics, Optimization, Differential Equations, Bayesian Networks, Markov Chains. |
| <b>Programming Languages and Tools</b>       | Python, Julia, R, MATLAB, Git, Docker, BASH, $\text{\LaTeX}$ , HTML.                                                   |

## EDUCATION

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|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024 - ...  | <b>PhD - Computer Science (Université Paris-Est Créteil)</b><br>Machine Learning of Continuous Systems Based on Probabilistic Approximations. Laboratory of Algorithms, Complexity, and Logic (LACL). |
| 2022 - 2024 | <b>Master's Degree - Mathematics, Modeling and Statistical Learning (Université Paris Cité)</b><br>Specialization: Modeling, Analysis, and Simulation. Graduated with High Honors.                    |
| 2020 - 2022 | <b>Bachelor's Degree - Fundamental and Applied Mathematics (Université Paris Cité)</b><br>Graduated with Honors.                                                                                      |
| 2019 - 2020 | <b>Double Bachelor's Degree - Mathematics and Computer Science (Université Paris Cité)</b><br>Completed first year with Honors.                                                                       |

## EXPERIENCE

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|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 02 - 08/2024 | <b>Research Engineer – Internship (Dassault Systèmes)</b><br>Research project on Deep Gaussian Processes for multi-fidelity simulation, with applications in thermal systems. Tasks included literature review, model development, experimentation, and results presentation via technical reports and seminars in the company's Research Division. |
| 06 - 10/2023 | <b>Administrative Assistant – University Contact Center (Université Paris Cité)</b><br>Assisted students with administrative procedures (enrollment, applications, diploma requests) through a contact center (phone and email support).                                                                                                            |

## PUBLICATIONS

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| 2026 | <b>Conference - ECMS 2026 (Grimstad, Norway)</b> <a href="#">[Paper]</a><br>A Sensitivity-Driven Sampling Reduction Method For Probabilistic Approximations of ODEs, O. Bouët-Willaumez, A. Le Coënt, B. Barbot, N. Pekergin, <i>International Conference on Modelling and Simulation (ECMS)</i> , 2026. |
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2026 **Conference - ROADEF 2026 (Tours, France)** [\[Paper\]](#)  
Sampling Methods for Probabilistic Approximations of ODEs, O. Bouët-Willaumez, A. Le Coënt, B. Barbot, N. Pekergin, *French Society of Operations Research and Decision Support (ROADEF), 2026.*

2025 **Conference - QEST + FORMATS 2025 (Aarhus, Denmark)** [\[Paper\]](#)  
Conservation Analysis and Discrete Probabilistic Approximations for Parameter Estimation of Biochemical Networks, O. Bouët-Willaumez, A. Le Coënt, B. Barbot, N. Pekergin, *Quantitative Evaluation of SysTems + Formal Modeling and Analysis of Timed Systems (QEST+FORMATS), 2025.*

## TEACHING

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2025-2026 **Website Design (Université Paris-Est Créteil)**  
First year of Computer Science Bachelor's Degree, 1st semester, 32h.

2025-2026 **Coding, Compression & Cryptography (Université Paris-Est Créteil)**  
Second year of Computer Science Bachelor's Degree, 2nd semester, 16h.

2025-2026 **Introduction to Complexity & Algorithms (Université Paris-Est Créteil)**  
Third year of Computer Science Bachelor's Degree, 2nd semester, 16h.

## PROJECTS

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2026 **ode2dbn-sensitivity** [\[Source Code\]](#)  
A Python sensitivity-guided framework for constructing reduced Dynamic Bayesian Networks from ODE models with efficient sampling of key dynamical dependencies.

2025 **NebulaNet** [\[Source Code\]](#)  
Python library for generating abstract SVG backgrounds using random point placement and proximity-based connectivity to produce constellation-like graph structures.

2025 **BayeSBML** [\[Source Code\]](#)  
Python library for conservation analysis and discrete probabilistic approximations to estimate parameters in biochemical networks.

## LANGUAGES

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**French** - Native, **English** - Fluent, **Spanish** - Basics.